



Black hole imaging

Find out how researchers produced the first ever image of a black hole.
<https://dgit.in/apr20-11>



Oldest reprogrammed cells

Scientists have managed to reprogram the stem cells of a 114-year-old woman.
<https://dgit.in/apr20-12>

equipment. Using the technology, scientists can quickly analyse samples, get the required data and store it for the future, all hands-free! The software also lets scientists communicate with peers across the world on their current projects. Add to this, audio and 2K visualisation abilities of models, samples and more, and you have an interactive workspace which fuels innovation.



Holo4Labs is a mixed reality solution which changes how scientists interact with lab equipment

TO THE SKIES

In outer worldly matters, NASA scientists have been tapping into the potential of VR to redefine our understanding of our galaxy and how it works. The scientists use a customised, 3D virtual reality simulation that brings to life the speed and direction of about 4 million stars in the Milky Way. Astronomer Marc Kuchner and researcher Susan Higashio stated that the virtual reality program dubbed PointCloudsVR, developed by Matthew Brandt, has helped them obtain a new perspective on stars' paths and improve their understanding of star groupings. The program could help correct some inconsistencies such as stars being classified in wrong groups as well as shed light on known star groups that may actually belong to larger groupings.

The findings derived from the program were presented at the annual American Geophysical Union (AGU) conference in December 2019. Kuchner and Higashio from NASA's Goddard Space Flight Center in Greenbelt, Maryland also plan to publish their findings next year after extensively using the novel program, along with engineer Matthew Brandt. As per NASA, Higashio stated that the associations between groups of stars "became more intuitive

inside the artificial cosmos found within the VR headset". The 3D visualisations help Goddard scientists understand how the local star neighbourhood was formed. The scientists said that using the program, they often discovered groups of young stars moving together which reveals that they could possibly be formed at the same time and same place. The PointCloudsVR software has

also been officially released and open sourced on NASA's Github: <https://dgit.in/PointCloudsVR>. Other star-tracking solutions as well as virtual hands-on applications for NASA engineers and other engineers are also developed which help with exploration and satellite servicing missions. Kuchner also believes that VR headsets

will become commonplace in scientific tools in the future and thinks of it as a research tool that should be present on every astrophysicist's desk.

NASA also utilises VR to train astronauts, where they are required to complete system rescue scenarios with VR technology. NASA's VRL (Virtual Reality Laboratory) is used, which is an immersive training facility providing real-time graphics and motion simulations. In the VRL astronauts also learn the basics of mass handling and robotics operations. NASA also utilises Charlotte, a VR robot built by McDonnell Douglas, which allows astronauts to feel the weight of an object in space.

FOR EDUCATIONAL PURPOSES

AR and VR are poised to transform STEM (Science, Technology, Engineering and Mathematics) curriculum and how it is taught. These immersive technologies can enhance learning through methods such as experiential learning through simulations, modelling and spatial representation of data, and the sense of presence and context. In 2017, the Stanford Human Computer Interac-

tion Lab released a new VR simulation dubbed The Stanford Ocean Acidification Experience which allows students to visit simulated oceans and experience the issue of ocean acidification. The project was developed in collaboration with Stanford marine biologists. Students utilised HTC Vives to notice the effect of CO₂ on marine life and even collect samples from the ocean floor.

VR technology is also used in The University of Michigan-Ann Arbor where VR 3D building models help better understand the structural integrity of engineering projects. Students as well as faculty members use the MIDEN (Michigan Immersive Digital Experience Nexus) technology for architectural walkthroughs, virtual reconstructions, and training scenarios for dangerous sites. Yet another example comes from a variety of projects conducted at Texas A&M where students study in the Immersive Mechanics Visualization Lab which depicts computational and experimental data in AR and VR environments. In this environment, 3D CAD models can be manipulated and tweaked in the development stages.

XR is even entering the classroom with software platforms such as Labster.



Labster offers a gamified experience to engage students

This platform has a suite of advanced lab simulations which encourage students to openly investigate and learn. The software even has a gamified experience which lets students embrace a CSI-type role of a forensics analyst who solves crimes within the lab. Additionally, these simulations also provide access to exorbitant equipment such as NGS (Next-Generation Sequencing) machines and electron microscopes that many universities cannot afford. 